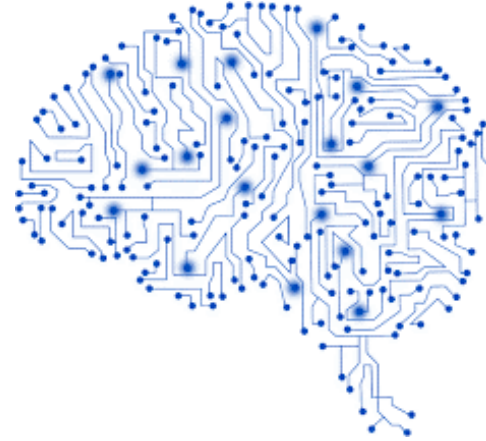




University of Minho
School of Engineering



Dados e Aprendizagem Automática

Intro to Data Science & Python/Scikit-learn

DAA 2025/2026

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Part I

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- Data Types
- Mean, Median & Mode
- Standard Deviation & Variance
- Probability Density Functions
- Percentiles
- Covariance & Correlation
- Virtual Environment
- Environment Setup
- Hands On



Data Types

Data Types

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- The most common data types:
 - Numerical
 - Categorical
 - Ordinal

Data Types

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Numerical

- It represents some sort of quantitative measurement
 - E.g., heights of people, page load times, and stock prices
- Discrete Data
 - It is integer-based; often represents the number of occurrences of an event
 - How many purchases did a customer make in a year?
 - How many times did I flip “heads”?
- Continuous Data
 - It can have an infinite number of values
 - How long did it take for a user to check out?
 - How much rain fell on a given day?

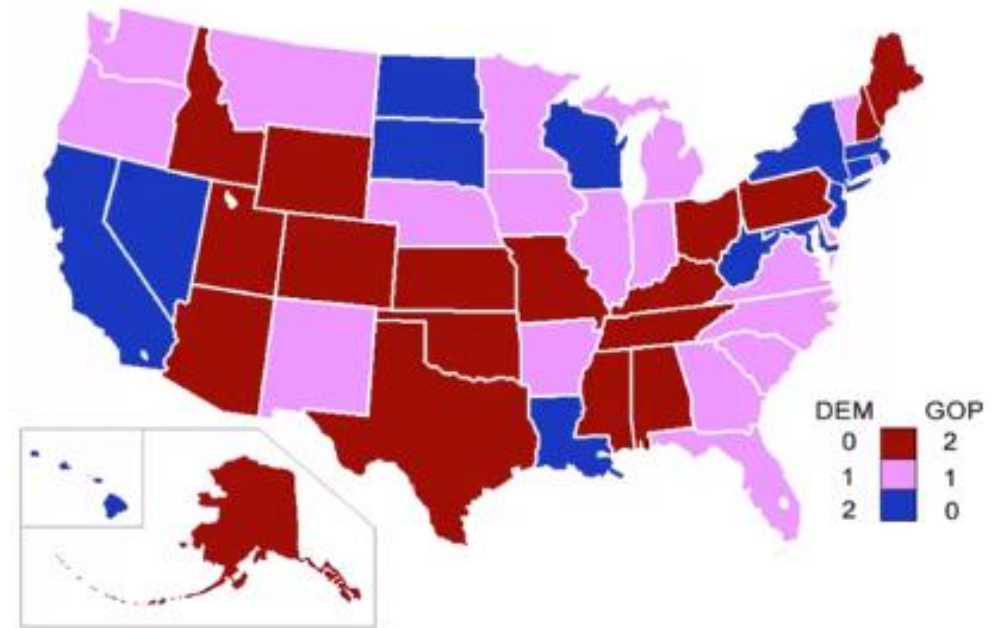


Data Types

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Categorical

- Qualitative data that has no inherent mathematical meaning
 - E.g., gender, yes/no (binary data), race, state of residence, product category, and political party
- Numbers can be assigned to these categories to represent them more compactly, but the numbers do not have any mathematical meaning



Data Types

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Ordinal

- A mixture of numerical and categorical data
- Categorical data with mathematical significance
- E.g., movie ratings on a scale of 1 - 5
 - Ratings must be 1,2,3,4 or 5
 - These values have a mathematical meaning: a rating of 1 means that the movie is worse than a movie with a rating of 2



Data Types

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Quick Quiz

- Classify the following types of data as numerical, categorical or ordinal.
 - How much gas/petrol is in your tank?
 - Rate your overall health on a scale of 1 – 4, where 1 is “poor”, 2 is “moderate”, 3 is “good” and 4 is “excellent”
 - The nationalities of your classmates
 - Ages in years
 - Money spent in a store





Mean, Median & Mode

Mean, Median & Mode

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Mean

- aka *average*
- Sum/number of samples
- Example:
 - The number of children in each house on a given street:

0, 2, 3, 2, 1, 0, 0, 2, 0

The MEAN is $(0+2+3+2+1+0+0+2+0)/9=\mathbf{1.11}$

Mean, Median & Mode

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Median

- Sort the values and select the one at the midpoint
- Example:

0, 2, 3, 2, 1, 0, 0, 2, 0

Sorting it:

0, 0, 0, 0, 1, 2, 2, 2, 3



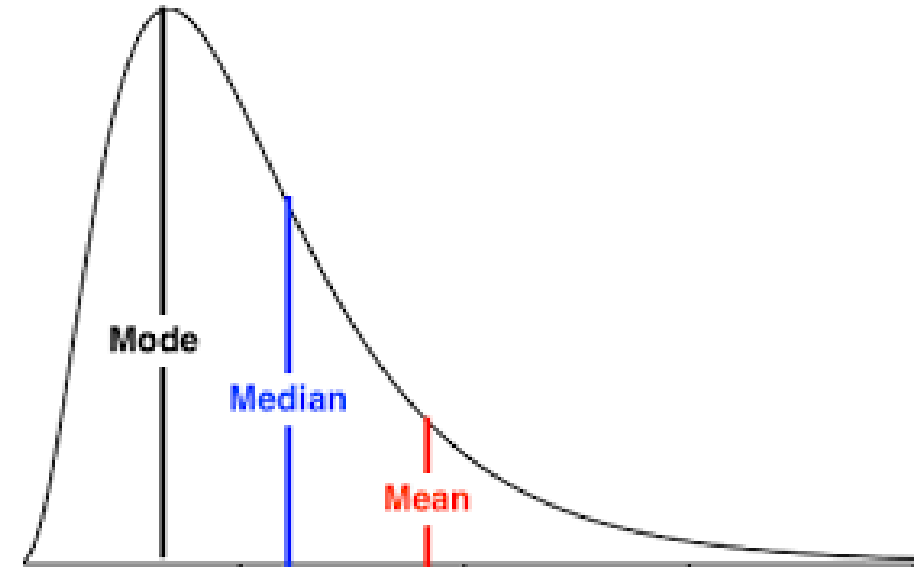
- If you have an even number of samples, take the average of the two in the middle.

Mean, Median & Mode

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Median

- The median is less susceptible to outliers than the mean
 - For example, the mean household income in the USA is \$72,641, but the median is only \$51,939 – because the mean is skewed by a handful of billionaires
 - In this example, the median better represents the “typical” American



Mean, Median & Mode

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Mode

- The most common value in a dataset
 - Not relevant to continuous numerical data
- The example of the number of children in each house:

0, 2, 3, 2, 1, 0, 0, 2, 0

How many of each value are there?

0: 4, 1: 1, 2: 3, 3: 1

The MODE is **0**

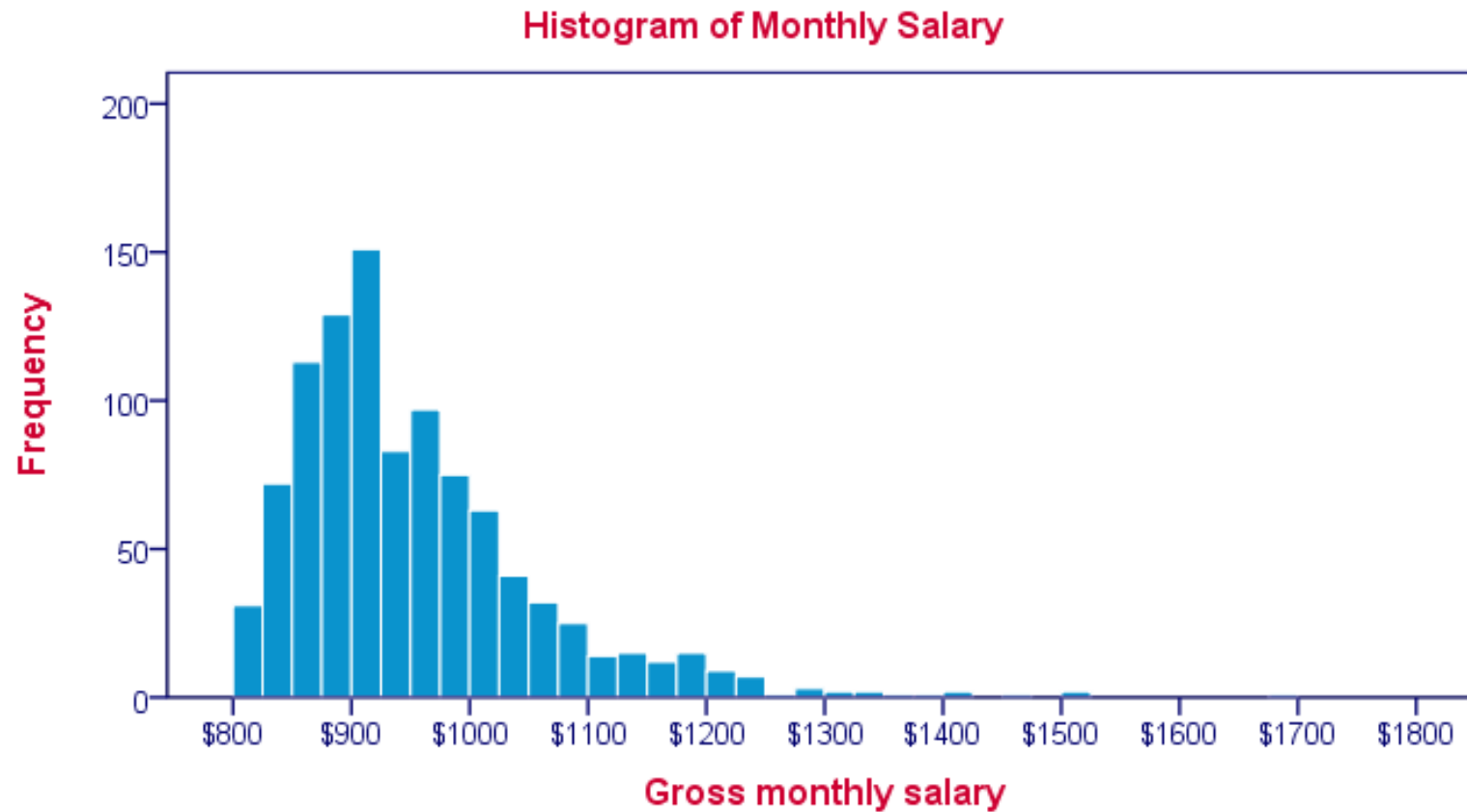


Standard Deviation & Variance

Standard Deviation & Variance

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Example of a histogram



Standard Deviation & Variance

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Variance measures how “spread-out” the data is

- Variance (δ^2) is simply the average of the squared differences from the mean
- Example:

What is the variance of the dataset **(1, 4, 5, 4, 8)**?

1. Find the mean: **(1+4+5+4+8) / 5 = 4.4**
2. Find the difference from the mean: **(-3.4, -0.4, 0.6, -0.4, 3.6)**
3. Find the squared differences: **(11.56, 0.16, 0.36, 0.16, 12.96)**
4. Find the average of the squared differences:

$$\delta^2 = (11.56+0.16+0.36+0.16+12.96) / 5 = \mathbf{5.04}$$

Standard Deviation & Variance

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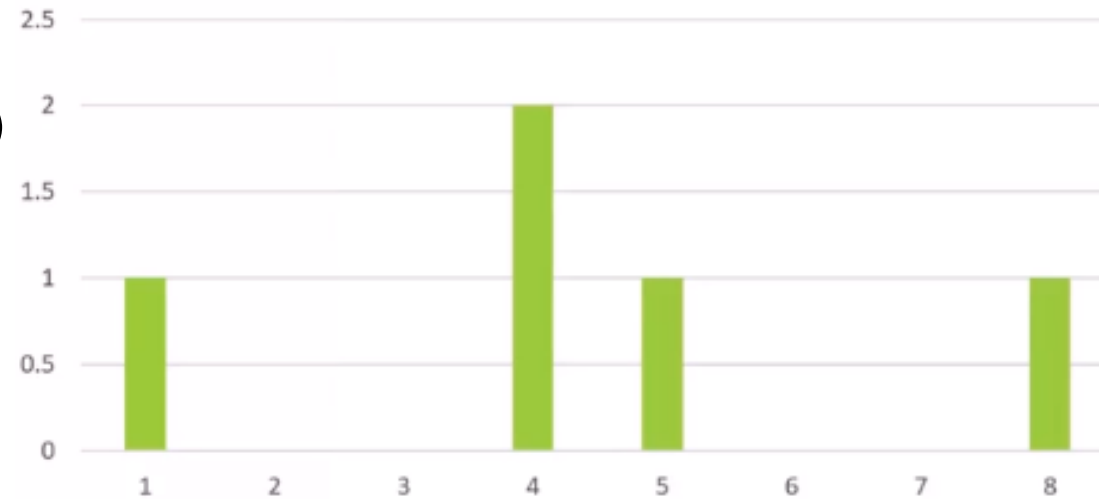
Standard Deviation, δ , is the square root of the variance

- It is usually used to identify outliers
- Data points lying more than one standard deviation away from the mean can be considered unusual
- The distance of a data point from the mean can be expressed in terms of “how many sigmas” it is away
- For example, given the case study = **(1,4,5,4,8)**

Mean = 4.4

$\delta^2 = 5.04$

$\delta = 2.24$





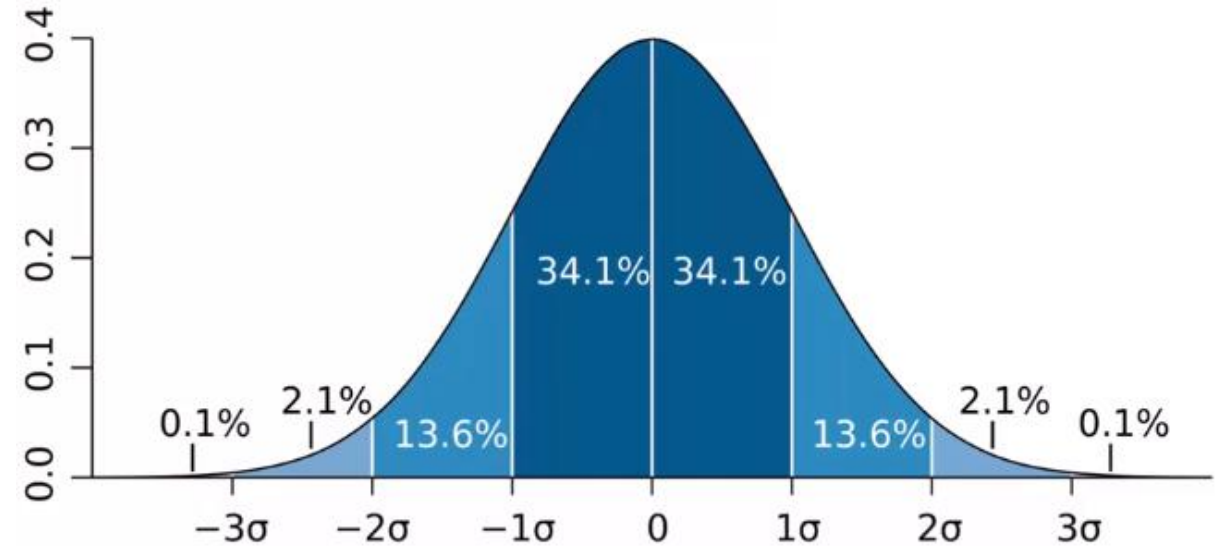
Probability Density Functions

Probability Density Functions

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“Normal Distribution”

- It gives the probability of a data point falling within a given range of a specific value
- A normal probability density function can be calculated based on histogram values

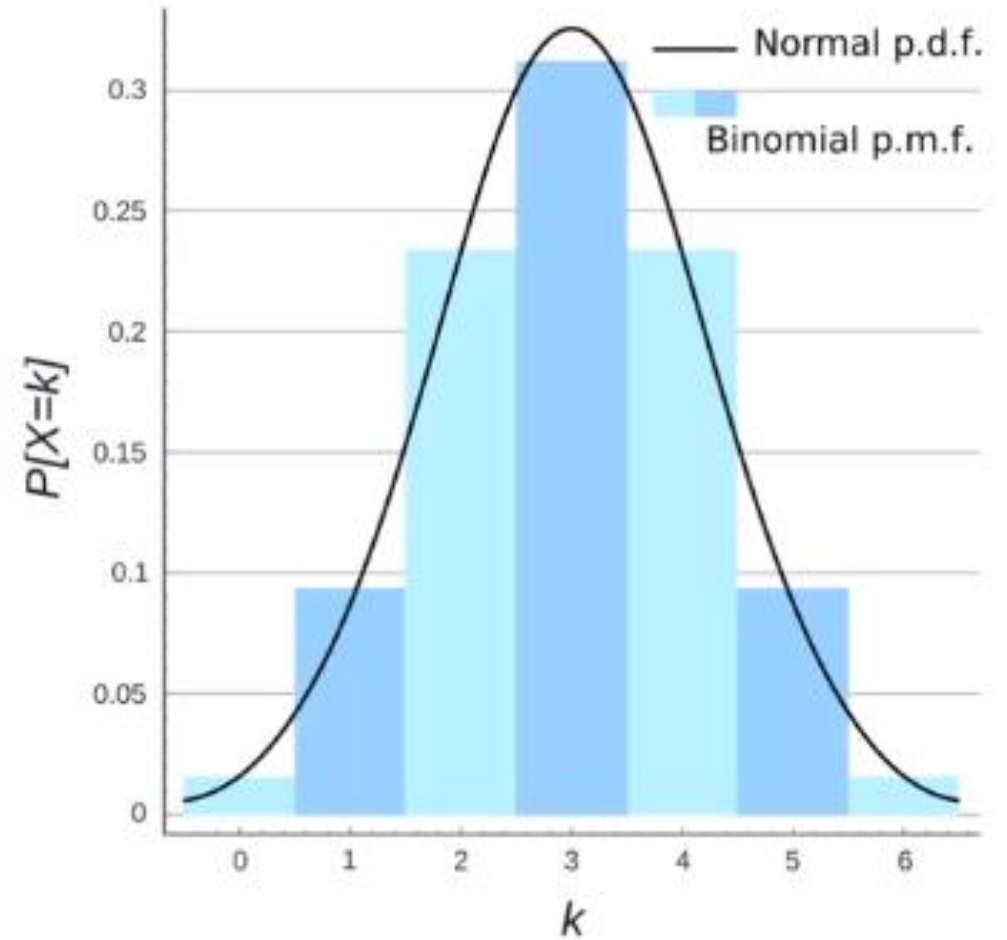


Probability Density Functions

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Probability Mass Function

- It is used for discrete data



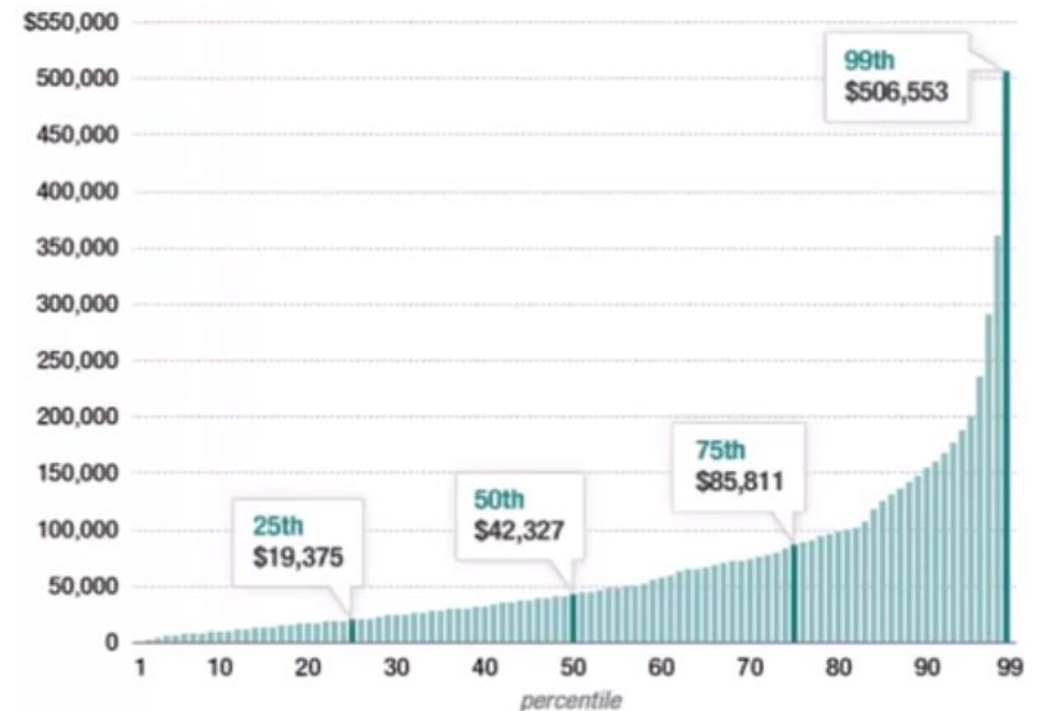


Percentiles

Percentiles

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- In a dataset, at what point are X% of the values less than that value?
- E.g., income distribution
 - Take all incomes from a country's population and sort them
 - The 99th percentile represents the income amount at which 99% of the population earns less than that value (i.e., \$506,553)

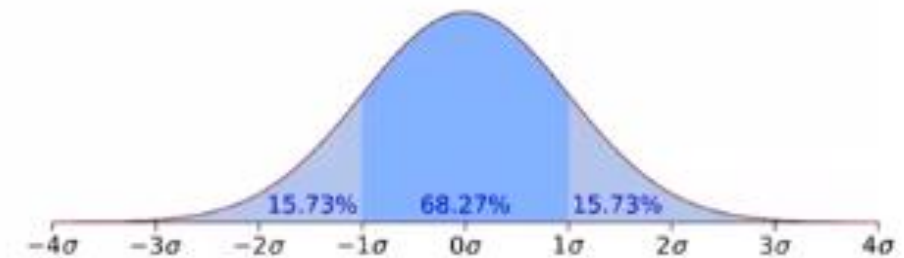
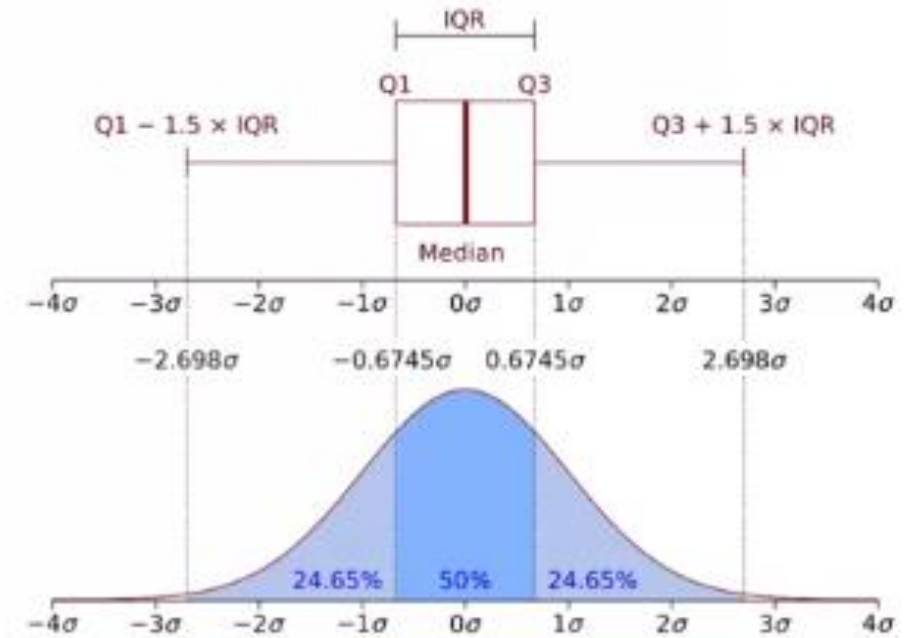


Percentiles

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Percentiles in a normal distribution

- The area between Quartile 1 and Quartile 3 represents 50% of the data distribution
- **IQR (Inter-Quartile Range)** represents the area in the middle of the distribution where the data is more concentrated





Covariance & Correlation

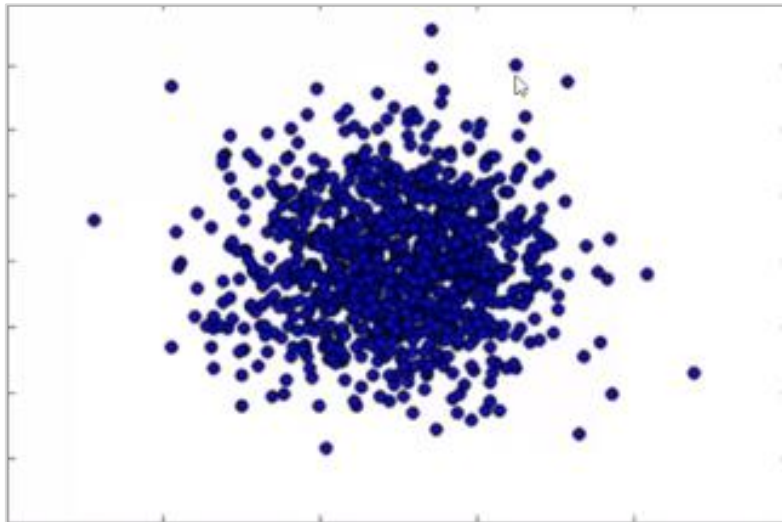
Covariance & Correlation

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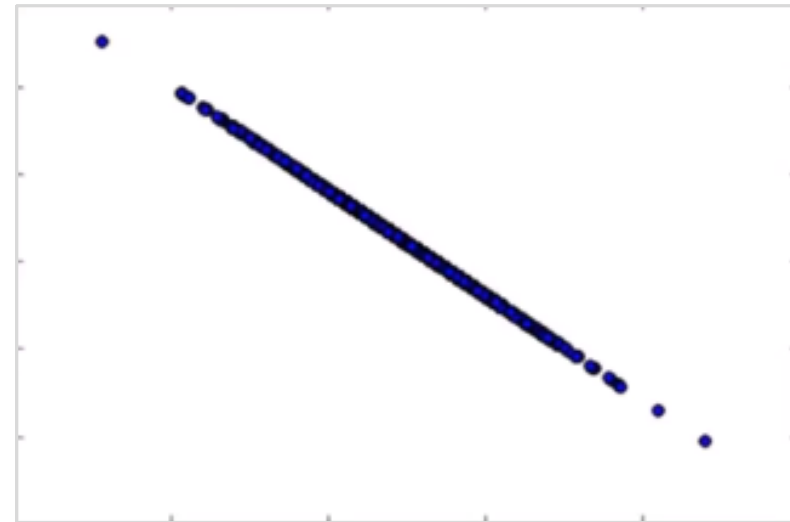
Covariance

- It measures how two variables vary in tandem from their means
- In other words, it shows how dependent are two attributes are on each other

Low Covariance



High Covariance



Covariance & Correlation

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Measuring it:

- Imagine the datasets for the two variables as high-dimensional vectors
- Convert these to vectors of variances from the mean
- Then, take the dot product (i.e., the cosine of the angle between them) of the two vectors
- Divide by the population size

Population Covariance Formula

$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

Sample Covariance

$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N-1}$$

Covariance & Correlation

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Interpreting **covariance** can be difficult

- A small covariance (close to 0) indicates a low degree of correlation between the two variables
- A large covariance (far from 0, and potentially negative for inverse relationships) indicates a correlation

Interpreting **correlation** is easier

- The normalization value is the covariance divided by the standard deviations of both variables
 - A correlation of -1 indicates perfect inverse correlation
 - A correlation of 0 indicates no correlation
 - A correlation of 1 indicates a perfect positive correlation

Covariance & Correlation

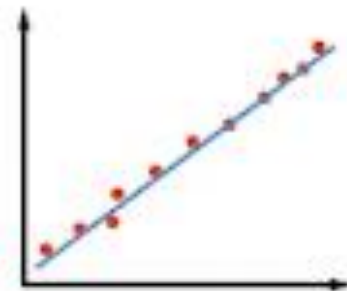
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Correlation does not imply causation!

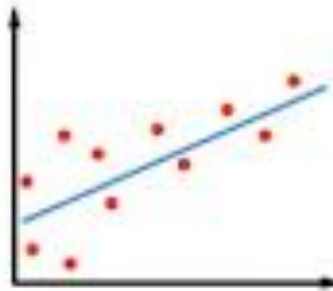
- Only a controlled, randomized experiment can provide insights into causation
- Use correlation to decide what experiments to conduct

Covariance & Correlation

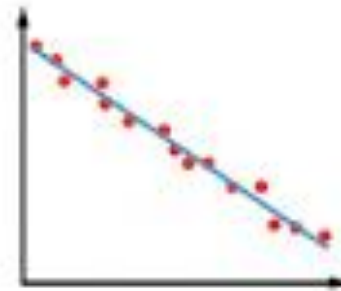
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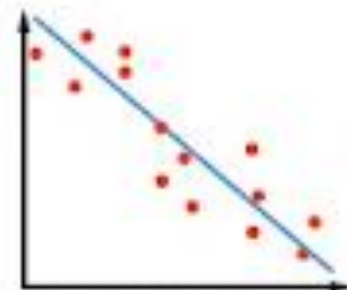
**STRONG POSITIVE
CORRELATION**



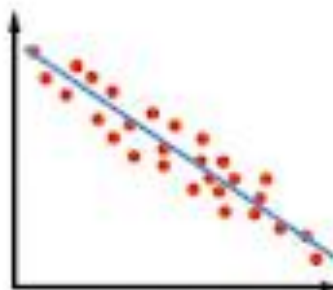
**WEAK POSITIVE
CORRELATION**



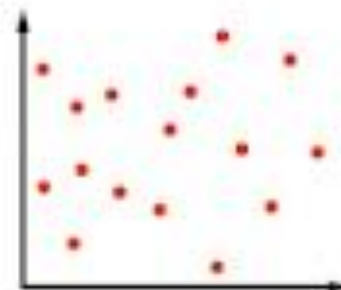
**STRONG NEGATIVE
CORRELATION**



**WEAK NEGATIVE
CORRELATION**



**MODERATE NEGATIVE
CORRELATION**



NO CORRELATION



Virtual Environments

Virtual Environments

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- Virtual environments allow to set up virtual installations of Python and its libraries on a local computer
- This means you can have multiple versions of Python and libraries, and easily activate or deactivate these environments
- Let's look at some examples of why you might want to do this

Virtual Environments

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- If you want to program in different versions of a library
- For example:
 - You develop a program with SciKit-Learn 0.17
 - Then, SciKit-Learn 0.18 is released
 - You want to explore the new version, but you don't want your old code to break
- If you may also want to ensure that your library installations are in the correct location
- For example:
 - You have multiple versions of Python on your computer
 - You might one environment with Python 2.7 and another with Python 3.6

- Conda has a built-in virtual environment manager that simplifies the entire process
- Since we don't require all of Conda's features, we suggest using **Miniconda** instead
- Check out the resource link for the official documentation:

<https://docs.conda.io/projects/miniconda/en/latest/>

- **Miniconda** is a free minimal Conda installer. It is a small bootstrap version of Anaconda that includes only Conda and Python, along with the packages they both depend on, and a few other useful packages (such as Pip and Zlib)
- If you require additional packages, you can use the `conda install` command to install from the thousands of packages available by default in Anaconda's public repository

- Command prompt usage example:

List all packages and versions installed	<code>conda list</code>
Create <i>myenv</i> environment	<code>conda create --name myenv python=3.13 numpy</code>
List environments	<code>conda info --envs</code>
Activate <i>myenv</i> environment	<code>conda activate myenv</code>
Enter Python	<code>python</code>
Import libraries	<code>import numpy as np</code> <code>import pandas as pd</code>
Exit Python	<code>quit()</code>
Installation of Pandas library	<code>conda install pandas</code>
Deactivate environment	<code>conda deactivate</code>

-> Error

Virtual Environments

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- **Python** also has a built-in virtual environment manager whose main advantage is that it simplifies library management across several projects
- However, it only includes the default Python packages, so all library installations are left up to the user

Environment creation	<code>python -m venv <venv_name></code>
Activation	macOS/Linux <code>. ./<venv_name>/bin/activate</code>
	Windows (PS) <code>.\<venv_name>\Scripts\Activate.ps1</code>
	Windows (cmd) <code>.\<venv_name>\Scripts\activate.bat</code>
Package installation	<code>pip install <package_name></code>
Environment deactivation	<code>deactivate</code>
Get package information	<code>pip freeze > requirements.txt</code>



Virtual Environments

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- **Windows Subsystem for Linux (WSL)** allows you to run a Linux environment in a Windows machine, enabling its features (Linux command-line tools, utilities and applications) without the need for a virtual machine or dual-booting
- How to install it:
 1. Open Windows PowerShell as an administrator
 2. Enter the following command: `wsl -install`
 3. Restart the computer
- Installation guides:
 - <https://learn.microsoft.com/en-us/windows/wsl/install>
 - <https://ubuntu.com/desktop/wsl>
- For Windows versions below 10 2004, see <https://learn.microsoft.com/en-us/windows/wsl/install-manual>



Environment Setup

Environment Setup

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- In this course we will use Jupyter Notebooks/Spyder for teaching purposes and to provide notes
- Please **note** that you are free to use **whichever development environment you prefer** (e.g., Spyder, PyCharm or VisualStudio)
- We will be using the Python 3.13 through the Miniconda distribution for this course

Environment Setup

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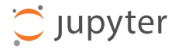
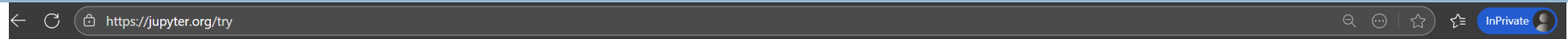
- If you are an experienced user who already has Python:
 - You may wish to install Jupyter and the required APIs using the Python package manager pip instead of Miniconda
 - Simply open your command prompt or terminal and enter the following command:

```
pip install jupyter
```

- We highly recommend that new users install Miniconda or Anaconda:
 - This will conveniently install Python, the Jupyter Notebook and other commonly used packages for scientific computing and data science
 - Visit www.jupyter.org to walk through the installation steps!

Environment Setup

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Try Jupyter

Use our tools without installing anything

Project Jupyter builds tools, standards, and services for many different use cases. This page has links to interactive demos that allow you to try some of our tools for free online, thanks to [mybinder.org](#), a free public service provided by the Jupyter community.

Applications

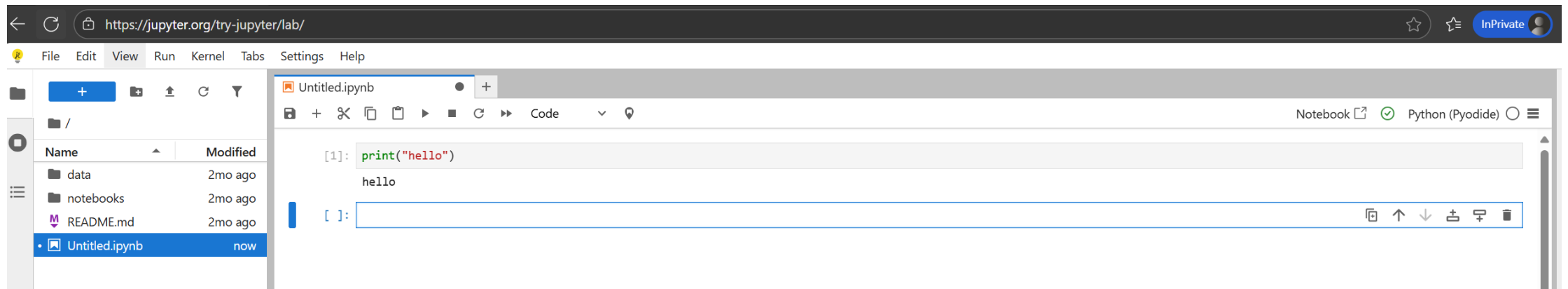
The Jupyter team builds several end-user applications that facilitate interactive computing workflows. Click the boxes below to learn how they work and to learn more. If you like one, you can find [installation instructions here](#).

⚠ Experimental ⚠ several of the environments below use the [JupyterLite project](#) to provide a self-contained Jupyter environment that runs in your browser. This is experimental technology and may have some bugs, so please be patient and report any unexpected behavior in [the JupyterLite repository](#).

JupyterLab	Jupyter Notebook	Voilà
		
The latest web-based interactive development environment	The original web application for creating and sharing computational documents	Share insights by converting notebooks into interactive dashboards

Environment Setup

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Hands On

Hands On

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T1

- We will use Scikit-learn/Sklearn
- Download and install the Miniconda package for your operating system (Windows, macOS, Linux) from <https://www.anaconda.com/docs/getting-started/miniconda/main>
 - Miniconda – Python 3.13
 - Deep Learning libraries are **not** required: Theano, Tensorflow, Keras
 - You will need to install Python (<https://www.python.org/downloads/>)
 - Setup guides (for reference):
 - <https://www.guvi.in/blog/how-to-setup-a-python-environment-for-machine-learning/>
 - <https://www.machinelearningplus.com/machine-learning/setup-python/>
 - <https://machinelearningmastery.com/setup-python-environment-machine-learning-deep-learning-anaconda/>

Hands On

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T2

- Start the Miniconda prompt and create a Python3.13 environment:
 - Open the terminal and execute:
`conda create --name envNAME python==3.13 numpy pandas matplotlib seaborn scikit-learn jupyterlab`
 - To install packages, enter the environment and execute:
`conda install PACKAGENAME`
 - To work inside the Python environment, execute:
`conda activate envNAME`
 - To exit the Python environment, execute:
`conda deactivate`

Hands On

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T2

- The following libraries must be installed in this environment:
 - Numpy
 - Pandas
 - Matplotlib
 - Seaborn
 - Scikit-learn
 - Jupyterlab

Hands On

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T3

- Activate the created virtual environment you created and check that the libraries have been installed
- Validate the installation of the library set presented in **T2**

T4

- Briefly check the documentation for each library mentioned in question **T2**
- Identify their relevance in the context of Machine Learning algorithm development